

Pricing Nonconvexities in Sequential Electricity Markets

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Outline

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Section 1. Pricing with nonconvexities

Pricing with nonconvexities

- ▶ Electricity markets are nonconvex due to start-up costs, minimum up and down times, and ramping constraints.
- ▶ Nonconvexities lead to the absence of competitive equilibrium.

Solution

- ▶ US markets: Efficient allocation with uplift payments
 - ▶ IP pricing (O'Neill et al., 2005)
 - ▶ Convex hull pricing (Hogan & Ring, 2003)
 - ▶ Consensus emerging towards CHP (Stevens et al., 2024)
 - ▶ Algorithms to compute them efficiently (Andrianesis et al. 2021; Stevens & Papavasiliou, 2022)
- ▶ EU market: Single price but allow paradoxically rejected bids (Madani & Van Vyve, 2014)

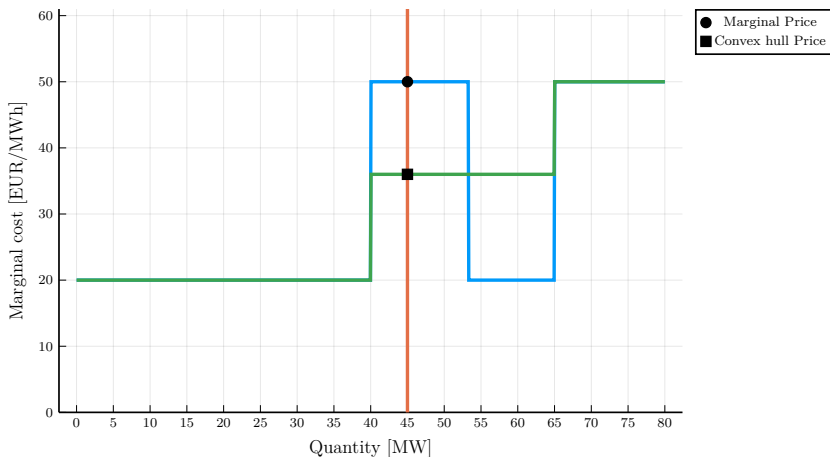
Example: Schiro et al. (2015)

	Generator 1	Generator 2	Generator 3
Min generation [MW]	0	25	0
Max generation [MW]	40	25	15
Marginal cost [€/MWh]	20	0	50
Start-up cost [€]	0	900	0

Two cases

- ▶ Case 1: Demand = 45 MW
 - ▶ $u^* = 0$
 - ▶ $\lambda^{MP} = 50$ €/MWh
 - ▶ Generator 2 prefers to commit
- ▶ Case 2: Demand = 55 MW
 - ▶ $u^* = 1$
 - ▶ $\lambda^{MP} = 20$ €/MWh
 - ▶ Generator 2 prefers not to commit

Two possible pricing methodologies



Theorem

Convex hull pricing minimize side payments ([Gribik et al., 2007](#))

Section 2. Sequential markets

Sequential markets

- ▶ Commitment and dispatch decisions are not made at the same time
- ▶ Day-ahead market: commitment decisions are decided (nonconvex), uncertainty about real-time conditions
- ▶ Real-time market: dispatch decisions can be updated (convex), no uncertainty

Studied in a series of papers by Mays and co-authors: [Mays \(2021\)](#); [Eldridge et al. \(2023a\)](#); [Eldridge et al. \(2023b\)](#); [Mays \(2024\)](#); [Santarém de Araújo et al. \(2026\)](#)

Stochastic unit commitment

$$\begin{aligned} \min_{u_g, p_g, c_g} \quad & \sum_{g \in G} \pi_\omega c_g(\omega) \\ \text{s.t.} \quad & \sum_g p_g(\omega) = D(\omega) \quad \forall \omega \in \Omega \\ & (u_g, p_g(\omega), c_g(\omega)) \in \mathcal{X}_g \quad \forall g \in G, \forall \omega \in \Omega \end{aligned}$$

Ex-post convex hull pricing

$$\begin{aligned} \min_{u_g, p_g, c_g} \quad & \sum_{g \in G} c_g(\bar{\omega}) \\ \text{s.t.} \quad & \sum_g p_g(\bar{\omega}) = D(\bar{\omega}) \\ & (u_g, p_g(\bar{\omega}), c_g(\bar{\omega})) \in \text{conv}(\mathcal{X}_g) \quad \forall g \in G \end{aligned}$$

Ex-ante convex hull pricing

$$\begin{aligned} \min_{u_g, p_g, c_g} \quad & \sum_{g \in G} \pi_\omega c_g(\omega) \\ \text{s.t.} \quad & \sum_g p_g(\omega) = D(\omega) \quad \forall \omega \in \Omega \\ & (u_g, p_g(\omega), c_g(\omega)) \in \text{conv}(\mathcal{X}_g) \quad \forall g \in G, \forall \omega \in \Omega \end{aligned}$$

Mays' framework

- ▶ Uplifts based on EP-CHP are too high because they include losses due to uncertainty
- ▶ Highlights that price formation needs to be considered in a sequential market context
- ▶ EA-CHP are impossible to compute
- ▶ Not an equilibrium
- ▶ Prices are set administratively
- ▶ Convex units need uplifts in real-time

Open question in EU setting

- ▶ What can be said about equilibrium when real-time prices are marginal prices ?

Section 3. Equilibrium

Equilibrium in sequential markets with risk neutral agents

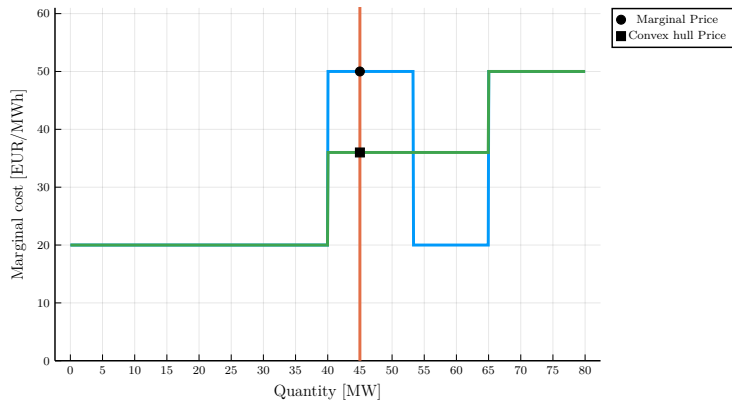
Conditions for equilibrium

- ▶ Real-time prices are marginal prices $\lambda^{MP}(\omega)$
- ▶ Participants make commitment decisions only based on $\lambda^{MP}(\omega)$
- ▶ These decisions should be consistent with real-time prices
- ▶ Non-anticipativity implies $\mathbb{E}[\lambda^{RT}] = \lambda^{DA}$

Implications

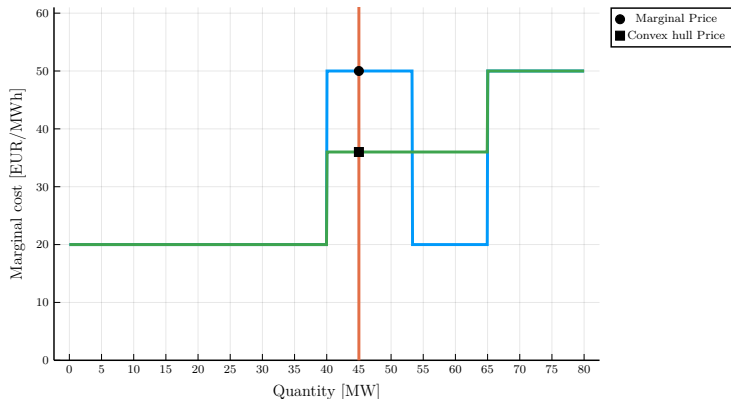
- ▶ **Weak price-taking assumption:** equilibrium can exist where it failed to exist before
- ▶ **Strong price-taking assumption:** equilibrium always exists in mixed-strategy

Weak price-taking assumption



→ $u^* = 0$ is optimal for generator 2 as it recognizes its impact on the price

Strong price-taking assumption



→ Randomized commitment $p = \frac{7}{15}$ is an equilibrium in mixed strategy for generator 2

Strong price-taking assumption

- ▶ Assume generators are described by $(MC_g, \bar{P}_g, SU_g)$
- ▶ Commitment incentive is given by:

$$\phi_g(\mu) = \bar{P}_g \mathbb{E}_{\lambda \sim \mu}[(\lambda - MC_g)^+] - SU_g.$$

- ▶ Equilibrium is defined by:

$$0 \leq \pi_g \perp -\phi_g(\mu) \geq 0$$

$$0 \leq 1 - \pi_g \perp \phi_g(\mu) \geq 0$$

$$\mu = \mu(\pi) = \sum_{u \in \{0,1\}^{\mathcal{N}}} \Pr(u \mid \pi) \delta_{\lambda(u)}$$

$$\Pr(u \mid \pi) = \prod_{g: u_g=1} \pi_g \prod_{g: u_g=0} (1 - \pi_g)$$

- ▶ Always exists by Kakutani's fixed point theorem

Section 4. Illustrative results

Case Study: set up

Instance

- ▶ Stylized Belgian system
- ▶ 62 generators (6 convex, 56 nonconvex)
- ▶ 8,760 hourly demand values, $D \in [3,495, 11,558]$ MW
- ▶ total capacity 15,260 MW
- ▶ VOLL = 5,000 €/MWh

(A) Strong price-taking equilibrium

- ▶ Set of fractional generators pre-computed based on LP relaxation
- ▶ Solve complementarity problem via Gauss–Seidel + bisection

(B) Euphemia (PAB forbidden, PAR allowed)

- ▶ Single MIP in primal-dual space, solved with Gurobi

Results

Total system cost (year, 8,760 hours):

Mechanism	Total cost	vs. optimal
Optimal (centralized MILP)	€2,063.8m	—
Two-settlement price-taking equilibrium	€2,104.3m	+1.96%
Euphemia (PAB forbidden / PAR allowed)	€2,187.6m	+6.00%

Section 5. Conclusion

Conclusion

Summary

- ▶ Accounting for the sequence of markets changes the conclusions of price formation with nonconvexities
- ▶ Focus should be set on real-time price formation, not day-ahead
- ▶ CHP vs MP discussion is replaced by administratively set vs equilibrium prices

Open questions

- ▶ More research to understand and characterize the equilibrium
- ▶ What can be done to drive this equilibrium (e.g. role of scarcity pricing in balancing)
- ▶ Accounting for risk aversion to understand the role of the day-ahead market price

Open positions at UCLouvain

- ▶ PhD position in Operations Research
- ▶ PhD position in Energy Economics
- ▶ TA position in Operations Research
- ▶ 2 open-rank professor positions in optimization and supply chain management

Thank You!

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